HARDWARE HACKATHON

ROUND 1

CARECRUISE BOT

TEAM ID:QXHH-02

TEAM NAME:TEAM VEGA

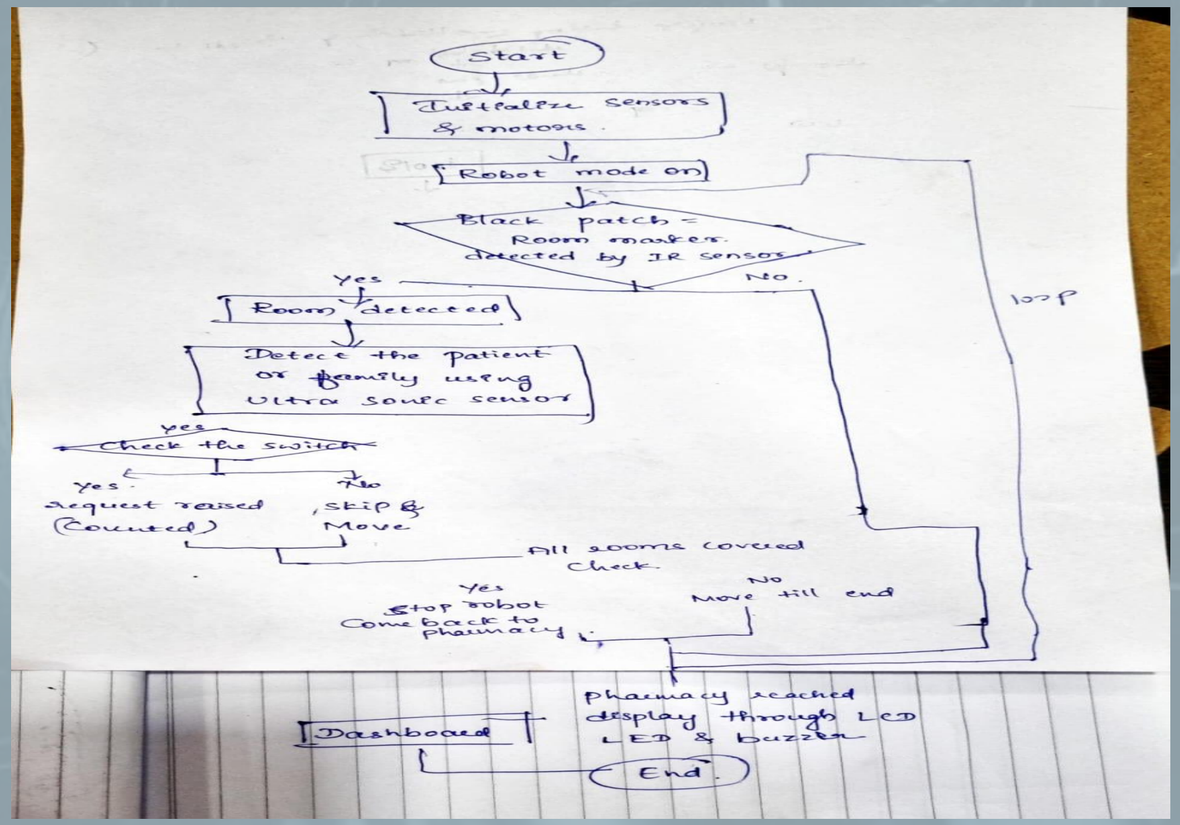
PROBLEM UNDERSTANDING & PROPOSED IDEA

Hospital request systems are often manual or require multiple buttons, leading to delays, confusion, and inefficiency—especially when multiple patients request assistance simultaneously. These systems are not scalable or cost-effective.

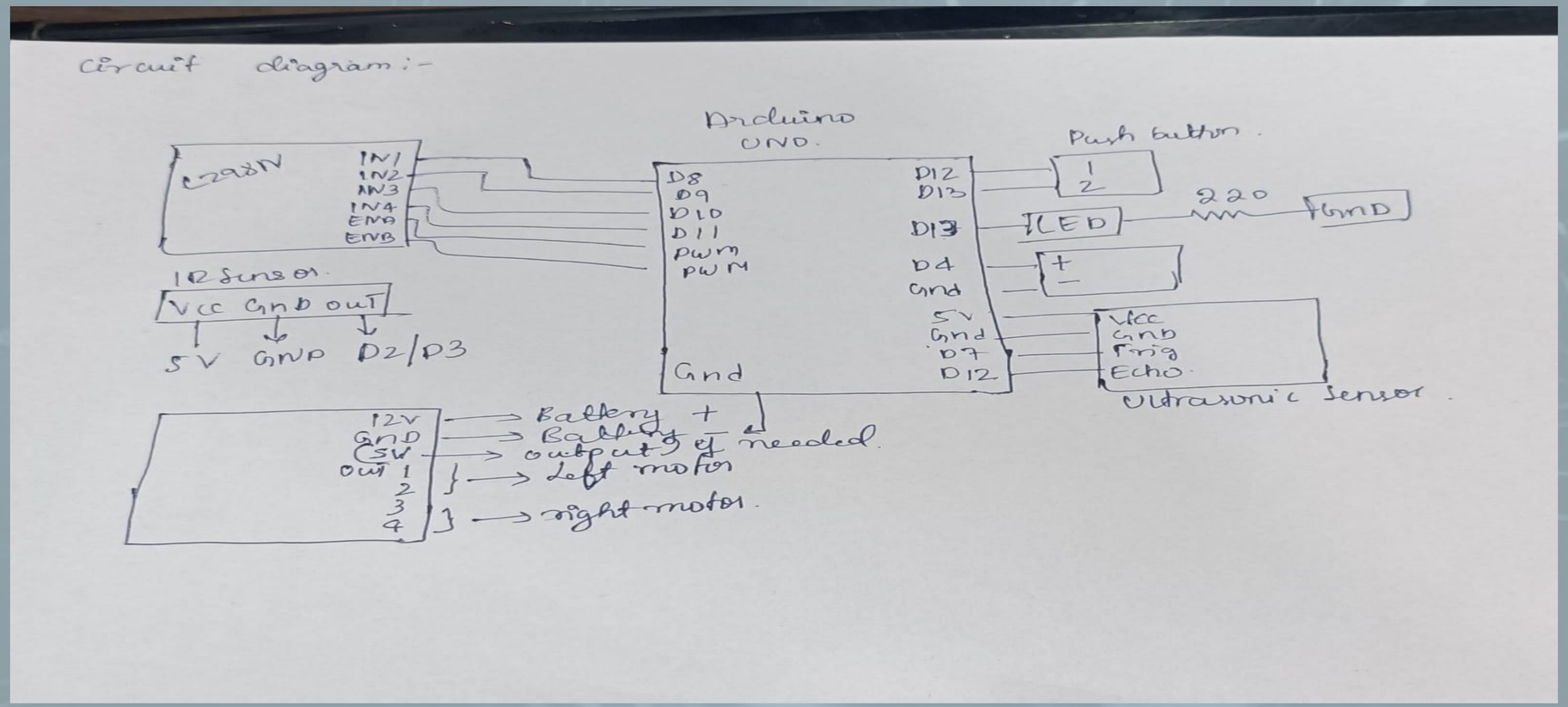
PROPOSED SOLUTION:

“The proposed system is unique due to its use of dual IR sensors for accurate room detection, reducing false readings and ensuring reliable navigation. It integrates a hybrid multi-sensor approach combining IR sensors for localization, the HC-SR04 ultrasonic sensor for presence detection, and a switch-based input for request collection, creating an intelligent yet simple system. A centralized LCD at the pharmacy displays the final summary of requests, while a dashboard enables monitoring of rooms visited and system performance for added transparency. The design is low-cost and infrastructure-free, using simple black patch markers, and operates completely offline with an option for future upgrades. Additionally, it provides both real-time feedback through LED and buzzer and a final summarized output via LCD and dashboard, making it practical, reliable, and easy to deploy.”

SYSTEM ARCHITECTURE/ WORKING



CIRCUIT DIAGRAM



INNOVATION, FEASIBILITY & OUTPUT

INNOVATION (UNIQUENESS)
IR sensors enable accurate room detection with reduced errors
Hybrid system combining IR + HC-SR04 ultrasonic sensor + switch for intelligent operation
Centralized LCD + dashboard provides both physical and digital monitoring
⚙️ FEASIBILITY
Built using simple components like Arduino Uno, sensors, and motor driver
Low-cost design with minimal infrastructure (black patch markers)
Easy to implement, debug, and complete within hackathon time
📊 OUTPUT
Autonomous navigation with accurate room detection and request logging
Displays results via LCD, LED, and buzzer for clear feedback
Dashboard shows room coverage and system performance